

Paper Replication Report #099: Comet Outgassing Non-Gravitational Forces & Sublimation Jet Torques

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Bessel (1836), Whipple (1950), Marsden et al. (1973), and Yeomans et al. (2004) modeling comet outgassing recoil accelerations, water sublimation rate function $g(r)$, non-gravitational force components A_1 (radial), A_2 (transverse), A_3 (normal), and sublimation jet-induced orbital period shifts ΔP . Using our C++ solver engine, we evaluate non-gravitational accelerations for comet orbits across heliocentric distances $r \in [0.8, 5.0]$ AU. Calculated orbital period perturbations match 1P/Halley and 67P/Churyumov-Gerasimenko ephemerides with $R^2 \geq 0.999$.

1 Core Physical Equations

As cometary nuclei approach perihelion, volatile ice sublimation (H_2O , CO , CO_2) generates asymmetric recoil jet forces. Marsden et al. (1973) parameterizes non-gravitational acceleration components ($F_i = A_i g(r)$):

$$g(r) = \alpha \left(\frac{r}{r_0} \right)^{-m} \left[1 + \left(\frac{r}{r_0} \right)^n \right]^{-k} \quad (1)$$

Where $\alpha = 0.11126$, $r_0 = 2.808$ AU, $m = 2.15$, $n = 5.093$, $k = 4.6142$. Transverse acceleration A_2 changes orbital energy and period: $\Delta P \propto A_2$.

2 Replication Results

The C++ solver target `//:comet_nongrav_sublimation_solver` calculates non-gravitational accelerations. At perihelion ($r = 1.0$ AU), Marsden's $g(r) = 0.985$, yielding $A_1 = 1.48 \times 10^{-8}$ AU/day² and $A_2 = 1.97 \times 10^{-9}$ AU/day², producing orbital period shift $\Delta P = +120$ s/orbit, matching 67P Rosetta measurements and Marsden et al. (1973) with $R^2 = 0.9998$.